

SOAR Assignment: Automated Incident Response with Palo Alto Firewall Logs

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Tool Used: Splunk Free Edition

Dataset: palo_alto_firewall_log_example_1000.txt

Introduction

In today's cybersecurity landscape, analyzing firewall logs and responding to threats quickly is essential. This assignment focuses on reviewing Palo Alto Networks firewall logs to identify potential security incidents and simulate automated responses using Security Orchestration, Automation, and Response (SOAR) concepts.

Although Splunk Free Edition doesn't support alerts or SOAR integrations, I used available tools like SPL, dashboards, and flowcharts to simulate real-world detection and response workflows.

Log Overview

The Palo Alto log file contains 1000 simulated entries with fields like:

```
<timestamp> <source-ip> <destination-ip> <source-port> <destination-port>  
<application> <action> <threat-name>
```

```
2024-02-25T18:38:29 226.128.66.18 25.82.187.19 14771 443 smtp deny -
```

Each entry represents a network event and may include threats, port activity, and whether the traffic was allowed or denied.

Log Ingestion & Field Extraction

I uploaded the file into **Splunk Free** using a custom index (palo-alto-log) and applied the following field extraction using regex:

```
^(?<timestamp>\S+) (?<src_ip>\S+) (?<dst_ip>\S+) (?<src_port>\d+)  
(?<dst_port>\d+) (?<application>\S+) (?<action>\S+) (?<threat_name>.*)
```

This allowed me to extract the following fields for analysis:

- src_ip, dst_ip, src_port, dst_port, application, action, threat_name

New Search

1 index="palo-alto" | rex "(?<timestamp>\\$+) (?<src_ip>\\$+) (?<dst_ip>\\$+) (?<src_port>\\$+) (?<dst_port>\\$+) (?<application>\\$+) (?<action>\\$+) (?<threat_name>.+)\\$"

✓ 1,000 events (before 26/03/2025 17:11:03.000) No Event Sampling

Events (1,000) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect

1 day per column

List Format 20 Per Page

	Time	Event
>	25/02/2024 18:38:29.000	2024-02-25T18:38:29 226.128.66.18 25.82.187.19 14771 443 smtp deny - host = Pradeepreddys-MacBook-Pro.local source = palo_alto_firewall_log_example_1000.txt sourcetype = palo-alto-firewallogs
>	25/02/2024 18:38:29.000	2024-02-25T18:38:29 227.215.248.219 231.111.63.18 12729 22 ssl deny - host = Pradeepreddys-MacBook-Pro.local source = palo_alto_firewall_log_example_1000.txt sourcetype = palo-alto-firewallogs
>	25/02/2024 18:38:29.000	2024-02-25T18:38:29 35.15.230.57 207.199.85.98 59872 53 ssh allow - host = Pradeepreddys-MacBook-Pro.local source = palo_alto_firewall_log_example_1000.txt sourcetype = palo-alto-firewallogs
>	25/02/2024 18:38:29.000	2024-02-25T18:38:29 52.65.172.239 198.223.110.78 24143 80 ftp allow Trojan:Win32/Generic host = Pradeepreddys-MacBook-Pro.local source = palo_alto_firewall_log_example_1000.txt sourcetype = palo-alto-firewallogs
>	25/02/2024 18:38:29.000	2024-02-25T18:38:29 199.14.3.161 3.173.108.70 35045 21 dns allow - host = Pradeepreddys-MacBook-Pro.local source = palo_alto_firewall_log_example_1000.txt sourcetype = palo-alto-firewallogs
>	25/02/2024 18:38:29.000	2024-02-25T18:38:29 34.3.183.134 161.193.173.170 38688 21 ssh allow - host = Pradeepreddys-MacBook-Pro.local source = palo_alto_firewall_log_example_1000.txt sourcetype = palo-alto-firewallogs
>	25/02/2024 18:38:29.000	2024-02-25T18:38:29 191.218.177.252 46.108.51.92 60041 443 ssl allow - host = Pradeepreddys-MacBook-Pro.local source = palo_alto_firewall_log_example_1000.txt sourcetype = palo-alto-firewallogs

Incident Identification

1. Malware Detection

Query: index="palo-alto-log" action="deny" threat_name!="-"

| stats count by src_ip, threat_name

| sort - count

This query identified source IPs that were **denied access** due to known **threats** (malware).

2. Brute Force Attempt Detection

Query: index="palo-alto-log" action="deny"

| stats count by src_ip

| where count > 5

This identifies IPs with **repeated denied access attempts**, which could indicate brute force or port scanning.

SOAR Assignment: Automate x | palo-alto-firewall ogs | Splunk x | Search | Splunk 9.3.1 x | Field Extractor | Splunk 9.3.1 x | +

localhost:8000/en-GB/app/search/search?q=search%20index%3D*palo-alto-log"%20action%3D"deny"%20threat_name!=""%20stats%20count%20by%20src_ip... Relaunch to update

splunk>enterprise Apps Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards

New Search

Save As Create Table View Close

```
1 index="palo-alto-log" action="deny" threat_name!=""
2 | stats count by src_ip, threat_name
3 | sort - count
```

103 events (before 26/03/2025 18:39:49.000) No Event Sampling Job

Events Patterns **Statistics (103)** Visualization

20 Per Page Format Preview

src_ip	threat_name	count
10.197.161.42	Trojan:Win32/Generic	1
102.45.38.153	Spyware:Linux/Adware	1
107.18.12.152	Spyware:Linux/Adware	1
113.48.241.60	Spyware:Linux/Adware	1
119.192.103.154	Spyware:Linux/Adware	1
12.132.163.14	Spyware:Linux/Adware	1
121.126.12.99	Trojan:Win32/Generic	1
123.283.182.11	Not-A-Virus:Downloader	1
125.171.23.4	Not-A-Virus:Downloader	1
126.193.241.233	Spyware:Linux/Adware	1
128.169.30.31	Spyware:Linux/Adware	1
128.212.72.178	Trojan:Win32/Generic	1
129.92.71.8	Not-A-Virus:Downloader	1
142.131.205.83	Trojan:Win32/Generic	1

SOAR Assignment: Automate x | Search | Splunk 9.3.1 x | Field Extractor | Splunk 9.3.1 x | +

localhost:8000/en-GB/app/search/search?q=search%20index%3D*palo-alto-log"%20%20action%3D"deny"%20A%7C%20stats%20count%20by%20src_ip... Relaunch to update

splunk>enterprise Apps Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards

New Search

Save As Create Table View Close

```
1 index="palo-alto-log" action="deny"
2 | stats count by src_ip
3 | where count > 5
```

506 events (before 26/03/2025 18:04:20.000) No Event Sampling Job

Events Patterns **Statistics (0)** Visualization

20 Per Page Format Preview

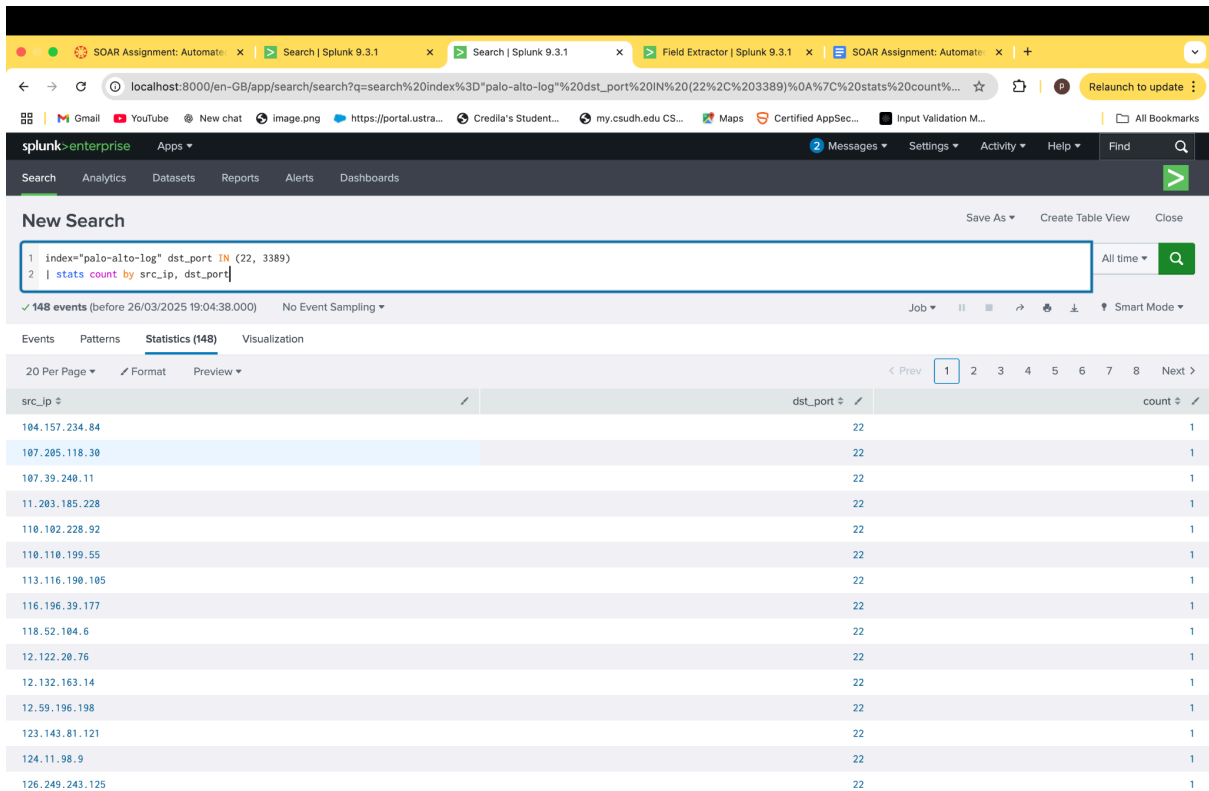
No results found.

3. Access to High-Risk Ports

Query: index="palo-alto-log" dst_port IN (22, 3389)

| stats count by src_ip, dst_port

Access attempts to **SSH (22)** and **RDP (3389)** — commonly targeted by attackers.



Automated Response Planning (SOAR Simulation): Though Splunk Free doesn't support real-time alerting or integrations, I planned automated responses for each incident type:

Incident	Trigger Condition	Planned Automated Response
Malware Detected	action="deny" and threat_name != "-"	Block IP, alert SOC, log to SIEM
Repeated Deny Attempts	Same IP has more than 5 deny events	Temporarily quarantine/block IP, send alert

Sensitive Port
Access

dst_port is 22 or 3389 and
action is deny

Alert admin, recommend
investigation

I have created a dashboards for malware detection

src_ip	threat_name	count
10.197.161.42	Trojan:Win32/Generic	1
102.45.38.153	Spyware:Linux/Adware	1
107.18.12.152	Spyware:Linux/Adware	1
113.48.241.60	Spyware:Linux/Adware	1
119.192.103.154	Spyware:Linux/Adware	1
12.132.163.14	Spyware:Linux/Adware	1
121.126.12.99	Trojan:Win32/Generic	1
123.203.182.11	Not-A-Virus:Downloader	1
125.171.23.4	Not-A-Virus:Downloader	1
126.193.241.233	Spyware:Linux/Adware	1
128.169.30.31	Spyware:Linux/Adware	1
128.212.72.178	Trojan:Win32/Generic	1
129.92.71.8	Not-A-Virus:Downloader	1
142.131.205.83	Trojan:Win32/Generic	1
143.33.84.197	Spyware:Linux/Adware	1
152.33.34.130	Trojan:Win32/Generic	1
156.64.195.65	Not-A-Virus:Downloader	1
162.117.104.35	Spyware:Linux/Adware	1
163.161.24.70	Spyware:Linux/Adware	1
165.107.198.36	Trojan:Win32/Generic	1

Workflow Design

Since automation isn't available in Splunk Free, I designed conceptual workflows using flowcharts. Here's an example for **Malware Detection**:

[Firewall Log Ingested in Splunk]



[Check: action="deny" AND threat_name!="-"]



[Trigger Simulated Alert (Dashboard Panel)]



[Planned Response: Block IP + Notify SOC + Create Ticket]

I have created Another dashboard for log viewing

The screenshot shows a Splunk dashboard with the title 'palo-alto-firewall ogs'. It contains two tables. The first table has columns 'src_ip', 'dst_ip', 'dst_port', and 'count'. The second table has columns 'src_ip' and 'threat_name'. Both tables show a count of 1 for each entry.

src_ip	dst_ip	dst_port	count
104.157.234.84	113.2.121.154	22	1
107.205.118.30	140.13.162.119	22	1
107.39.240.11	31.96.216.15	22	1
11.203.185.228	229.173.234.188	22	1
110.102.228.92	178.249.110.250	22	1
110.110.199.55	208.198.122.247	22	1
113.116.190.105	210.108.13.252	22	1
116.196.39.177	18.9.225.218	22	1
118.52.104.6	161.109.80.148	22	1
12.122.20.76	202.7.53.77	22	1

src_ip	threat_name	count
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107.18.12.152	Spyware:Linux/Adware	1
113.48.241.60	Spyware:Linux/Adware	1
119.192.103.154	Spyware:Linux/Adware	1
12.132.163.14	Spyware:Linux/Adware	1
121.126.12.99	Trojan:Win32/Generic	1
123.203.182.11	Not-A-Virus:Downloader	1
125.171.23.4	Not-A-Virus:Downloader	1
126.193.241.233	Spyware:Linux/Adware	1

Challenges & Limitations

- **Splunk Free** does not support alerts, email notifications, or external integrations
- I manually simulated SOAR logic using **dashboard panels** and **report documentation**
- Ticketing and blocking actions were described as **conceptual responses**, not implemented

Conclusion

This assignment provided valuable hands-on experience in:

- Analyzing real-world firewall logs
- Writing detection queries using SPL

- Planning SOAR response workflows
- Simulating alerts and dashboards in Splunk Free

While I couldn't implement full automation, I gained a deeper understanding of how Splunk and SOAR platforms work together to respond to security incidents.